Innovation in UX/UI

Tech Laboratory 2

What made the difference?



2020

GPT-3 becomes available through an API. Developers are impressed. Almost nobody else notices.



2022

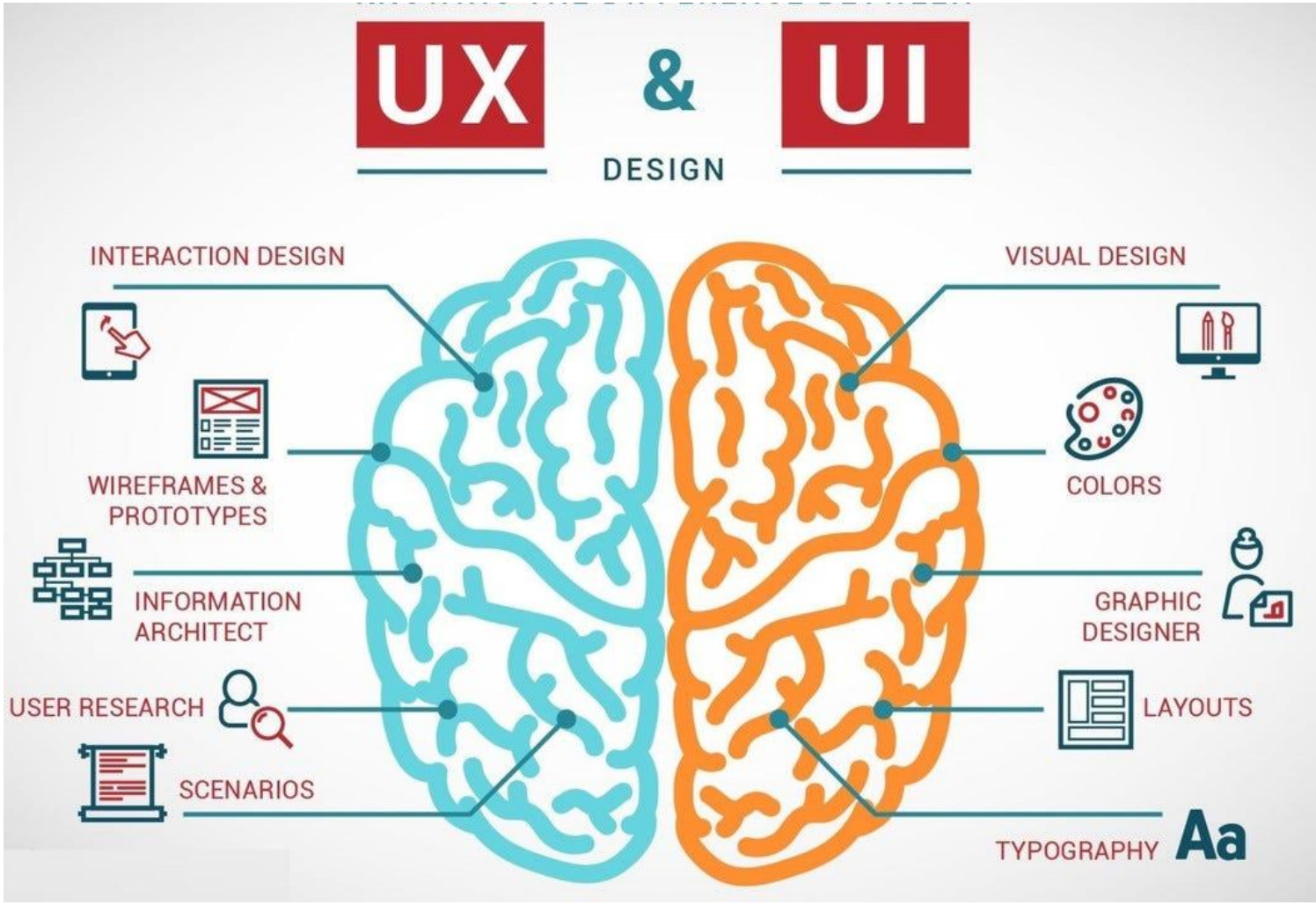
ChatGPT: a model from the same family + a chat box. 100 million users in 2 months.
(TikTok needed 9 months. Instagram, 2.5 years.)

What most people saw as the breakthrough was the interface.
That is UX.

UBS analyst note, reported by Reuters (Feb. 2023): ChatGPT reached ~100 million monthly users in January 2023, two months after launch.

UX vs UI

- **UX:** the whole experience of using the product
  - Strategy & content
  - Wireframes & prototypes
  - Testing & analysis
- **UI:** what you see and touch
  - Look & feel
  - Responsiveness & interactivity



*https://miro.medium.com/v2/resize:fit:1050/1*hp-yfKsmzsj711iLbM8eEw.jpeg

Understanding Good UX: An Important Truth



Image: https://webdesignledger.com/35-design-jokes-will-make-you-laugh/

Jakob Nielsen
Holds 79 US patents, mostly on making the web easier to use.

Don Norman
Former VP of Advanced Technology at Apple. He coined the term “user experience”.

10 Usability Heuristics for User Interface Design

Nielsen Norman Group · first published in 1990, refined in 1994

**https://www.nngroup.com/articles/ten-usability-heuristics/*



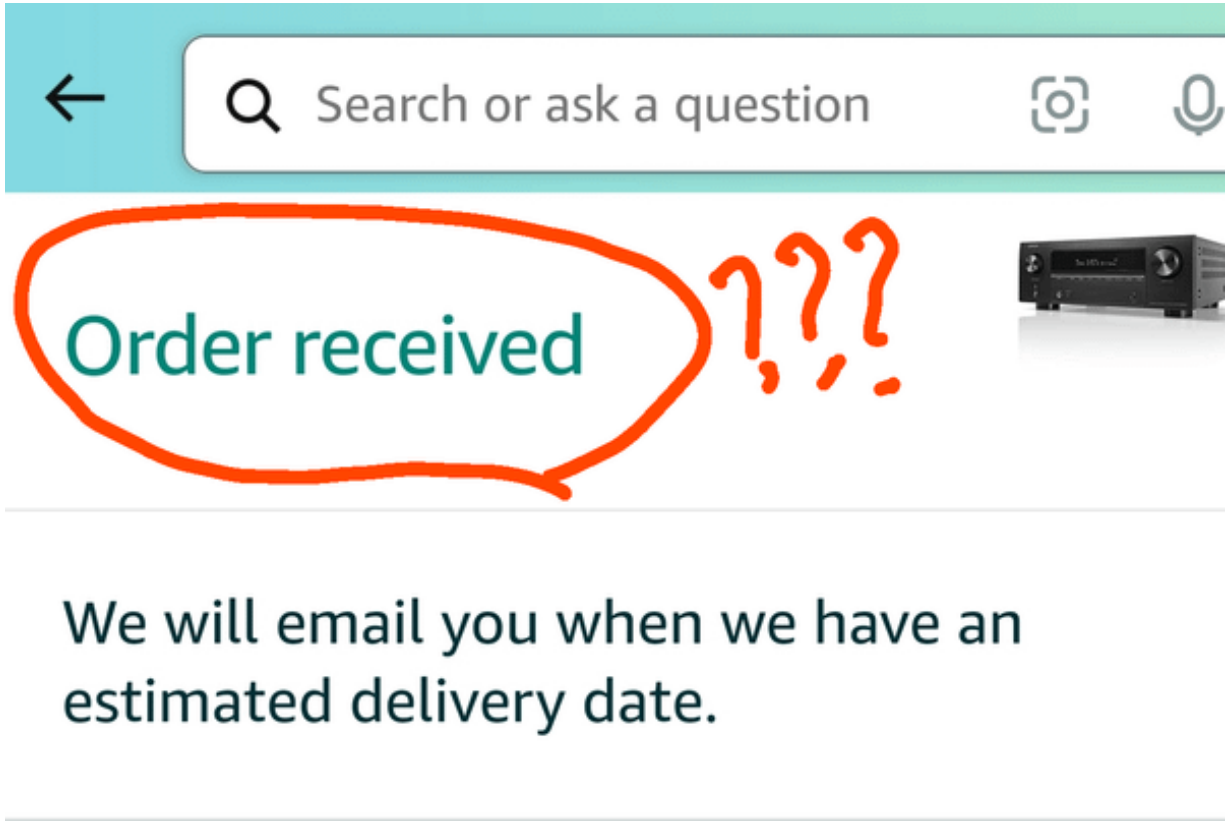
Let's Play a Game: Good UX vs. Bad UX

Theme 1: The Foundation - Clarity & User Confidence

Heuristics #1, #2, and #4

<- Left or right ? ->



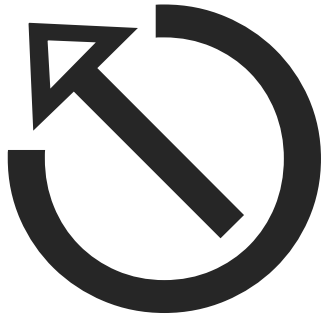


Heuristic #1: Visibility of System Status

Great UX always keeps the user informed.

Which icon do you understand immediately without thinking?

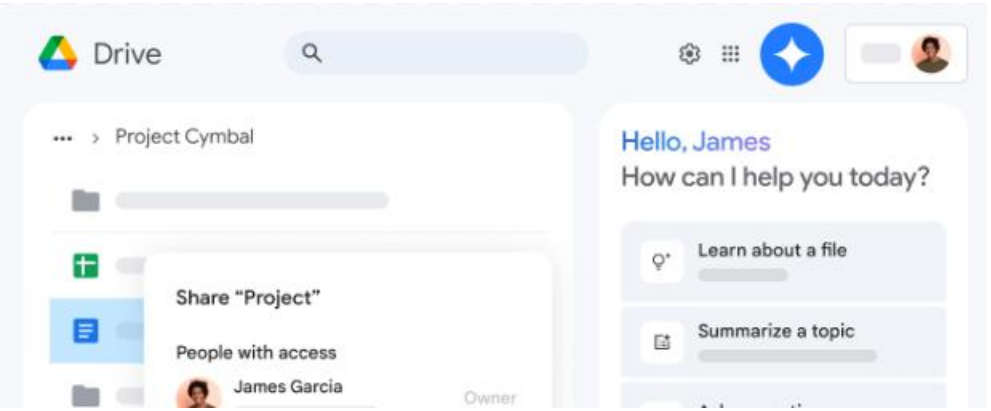


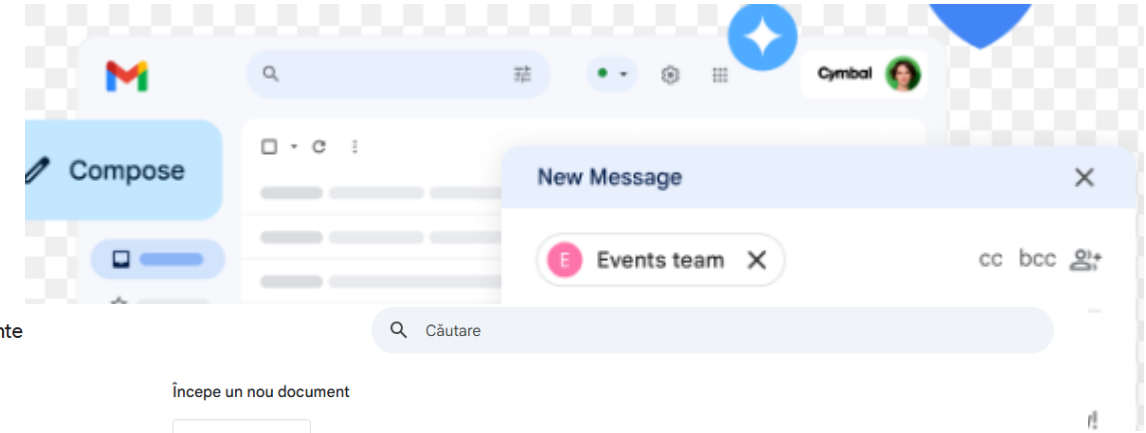


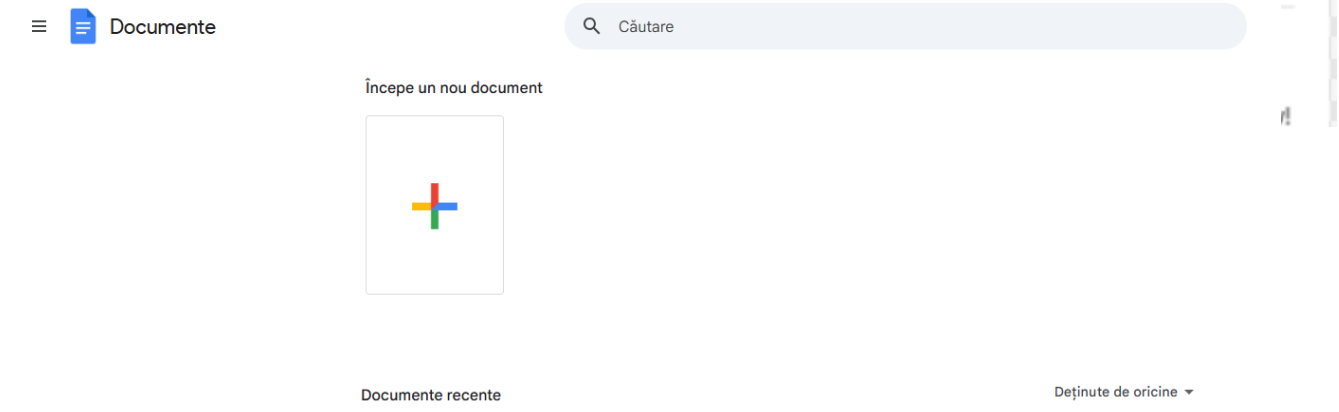
#2: Match Between System and the Real World

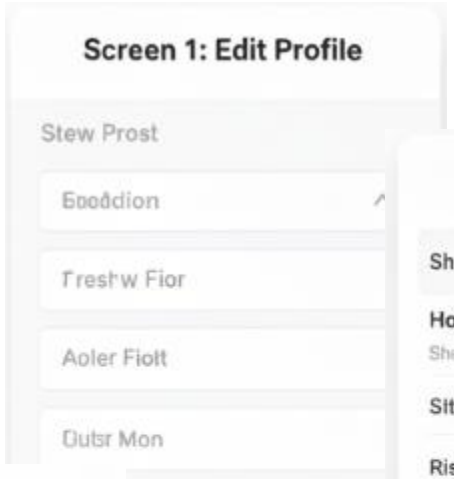
The system should speak the users' language and use familiar conventions.

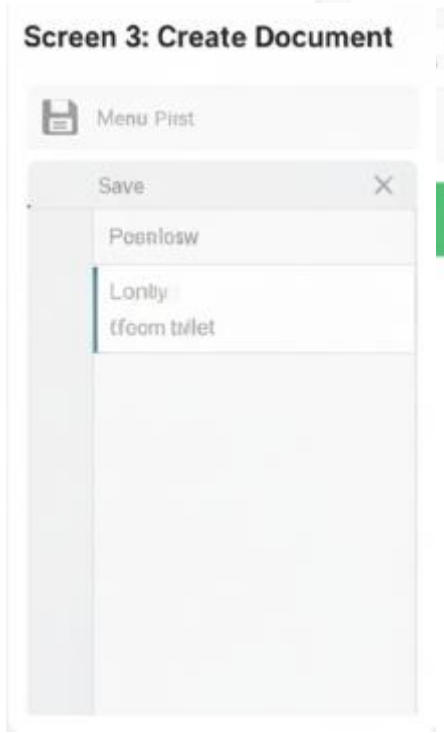
The Power of Patterns



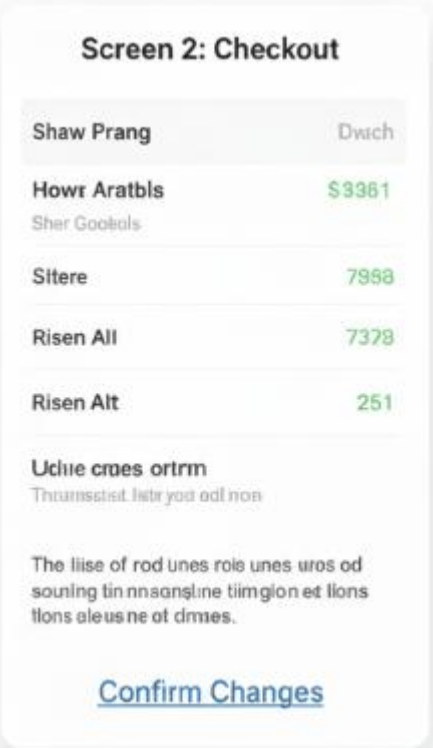












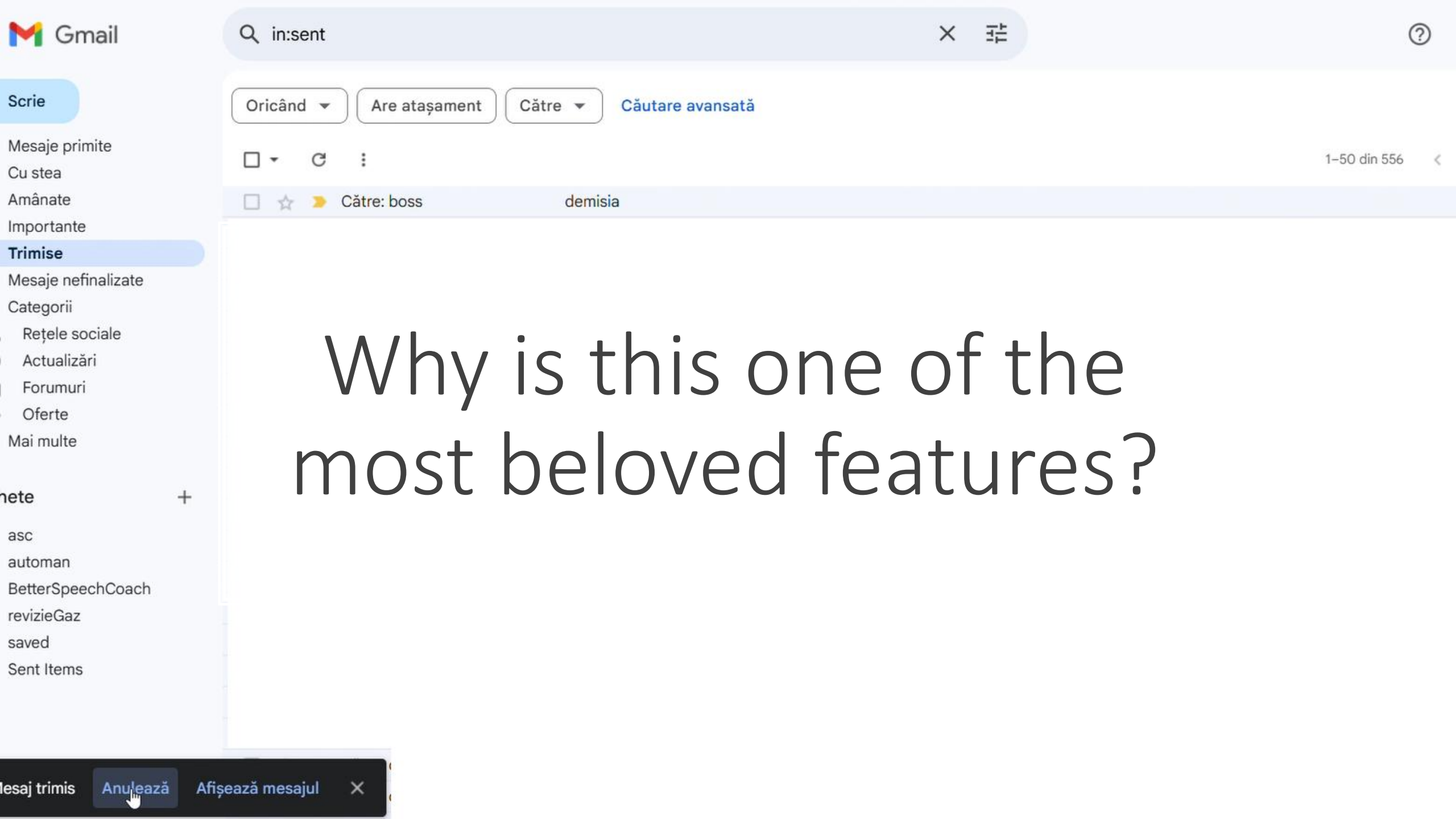
Heuristic #4: Consistency and Standards

Users should never have to wonder if different words, icons mean the same thing

- Internal Consistency: The Google Workspace
- External Consistency: following platform and industry conventions - green for environment, blue for airplane

Theme 2: The Safety Net - Forgiveness & Error Handling

Heuristics #3, #5, #9 and #10



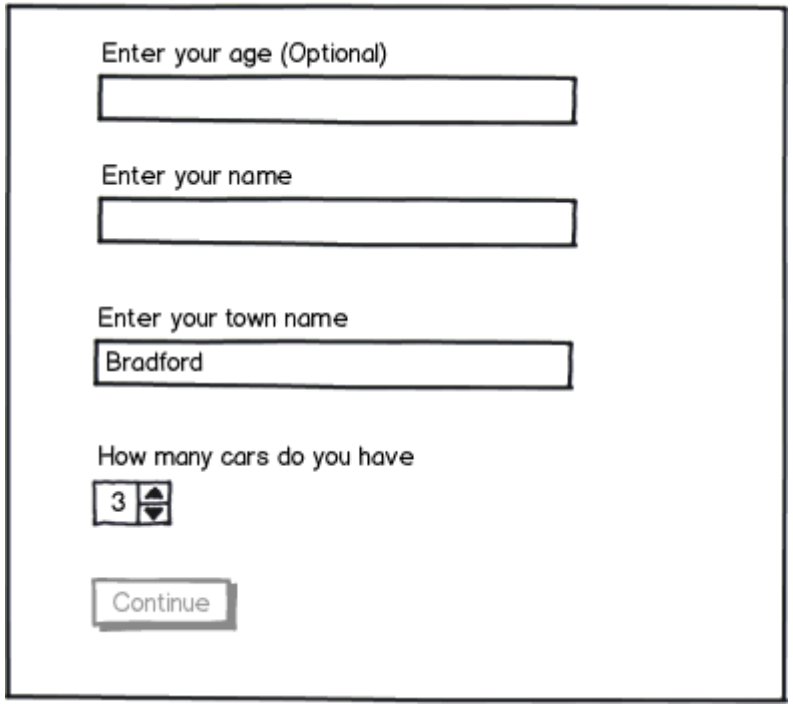
Why is this one of the most beloved features?

Heuristic #3: User Control and Freedom

Users need a clear 'emergency exit' to leave an unwanted state

'Undo' is the ultimate safety net.

Error Prevention vs. Recovery





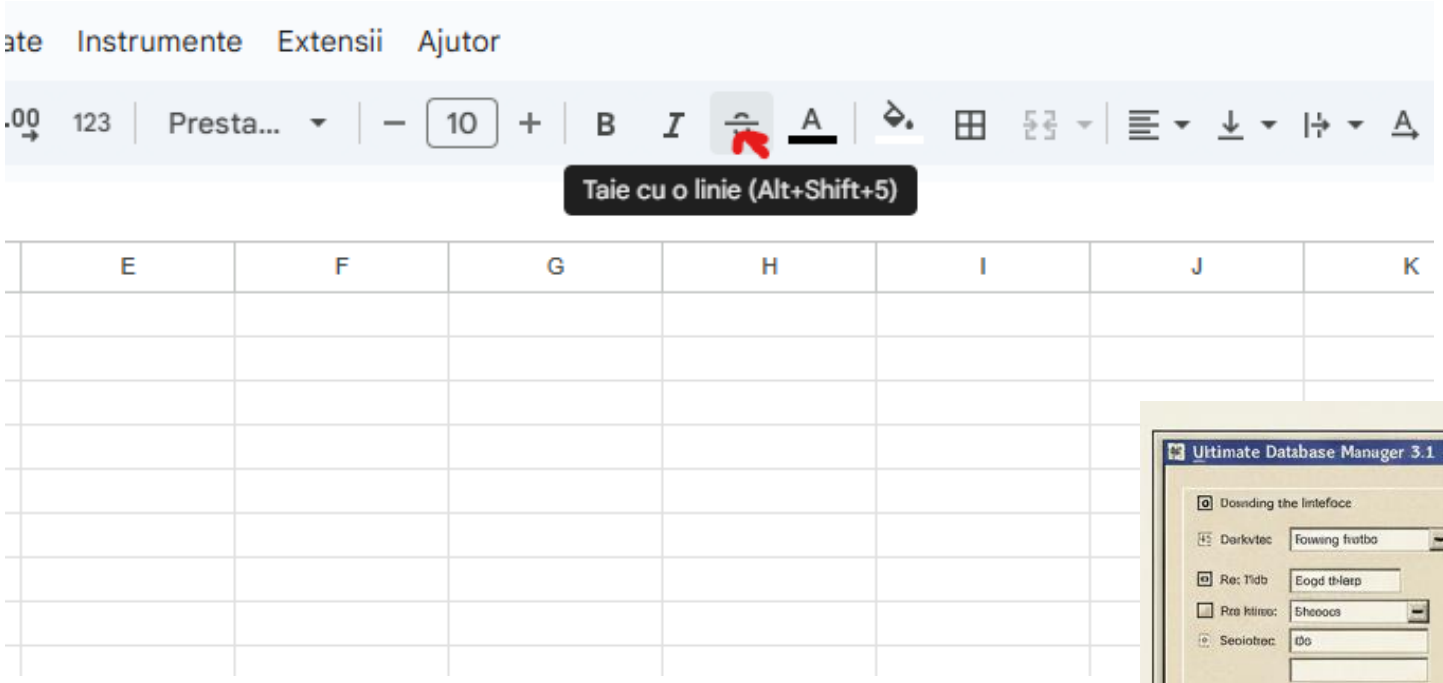
Heuristic #5: Error Prevention

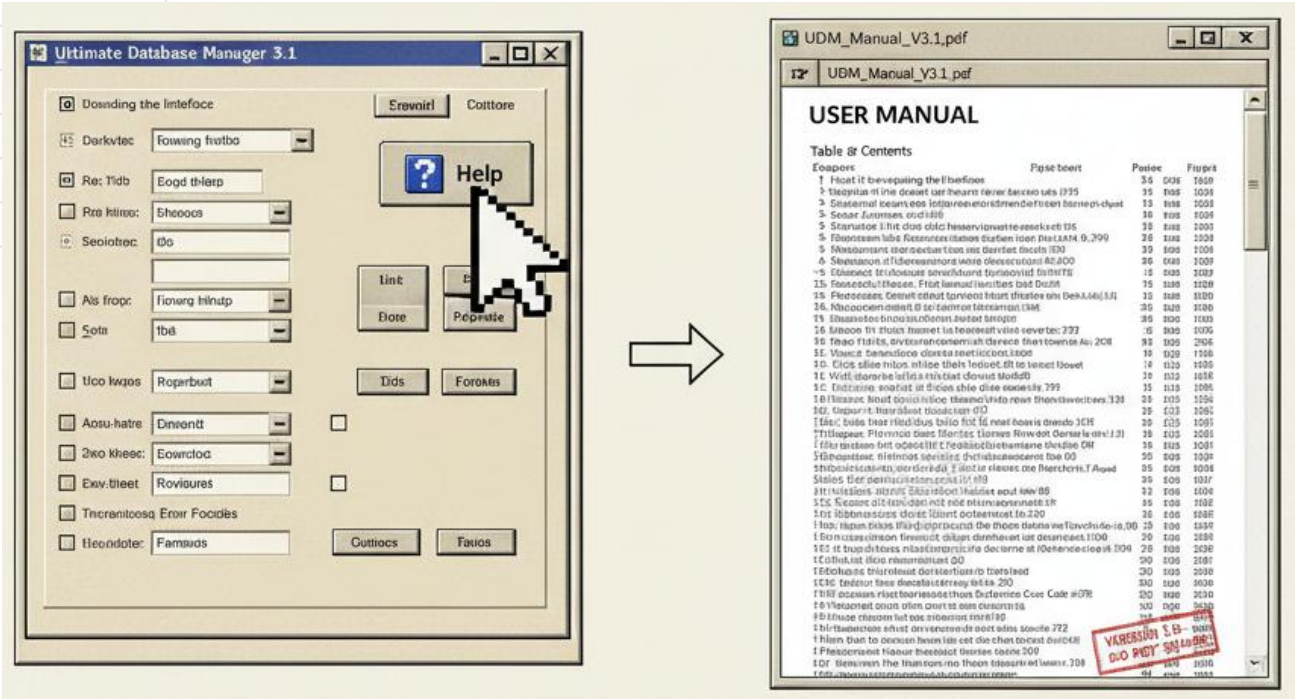
making it impossible to fail

Heuristic #9: Help Users Recover from Errors

making the failure state less
frustrating, understandable and even fun

When All Else Fails, Read the (Right) Manual!





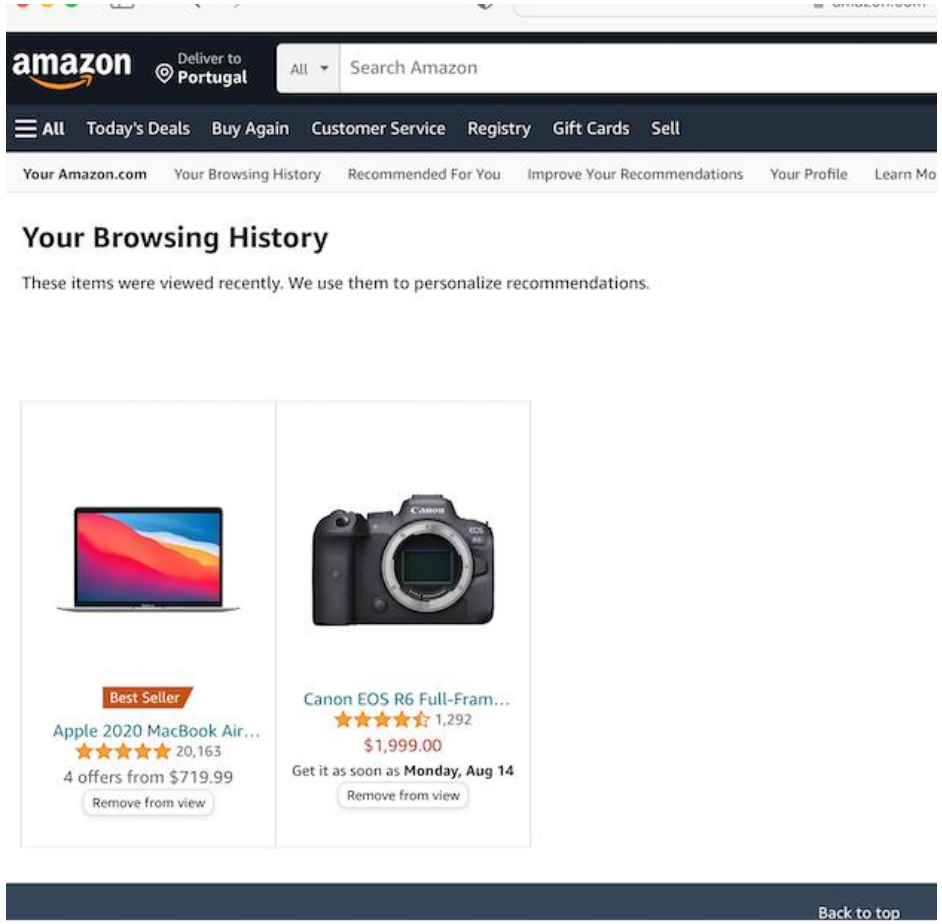
Heuristic #10: Help and Documentation

Even the best systems need good help, but *how* that help is delivered makes all the difference

Theme 3: The Accelerator - Efficiency & Elegance

Heuristics #6, #7, and #8

Why are those sections so effective?

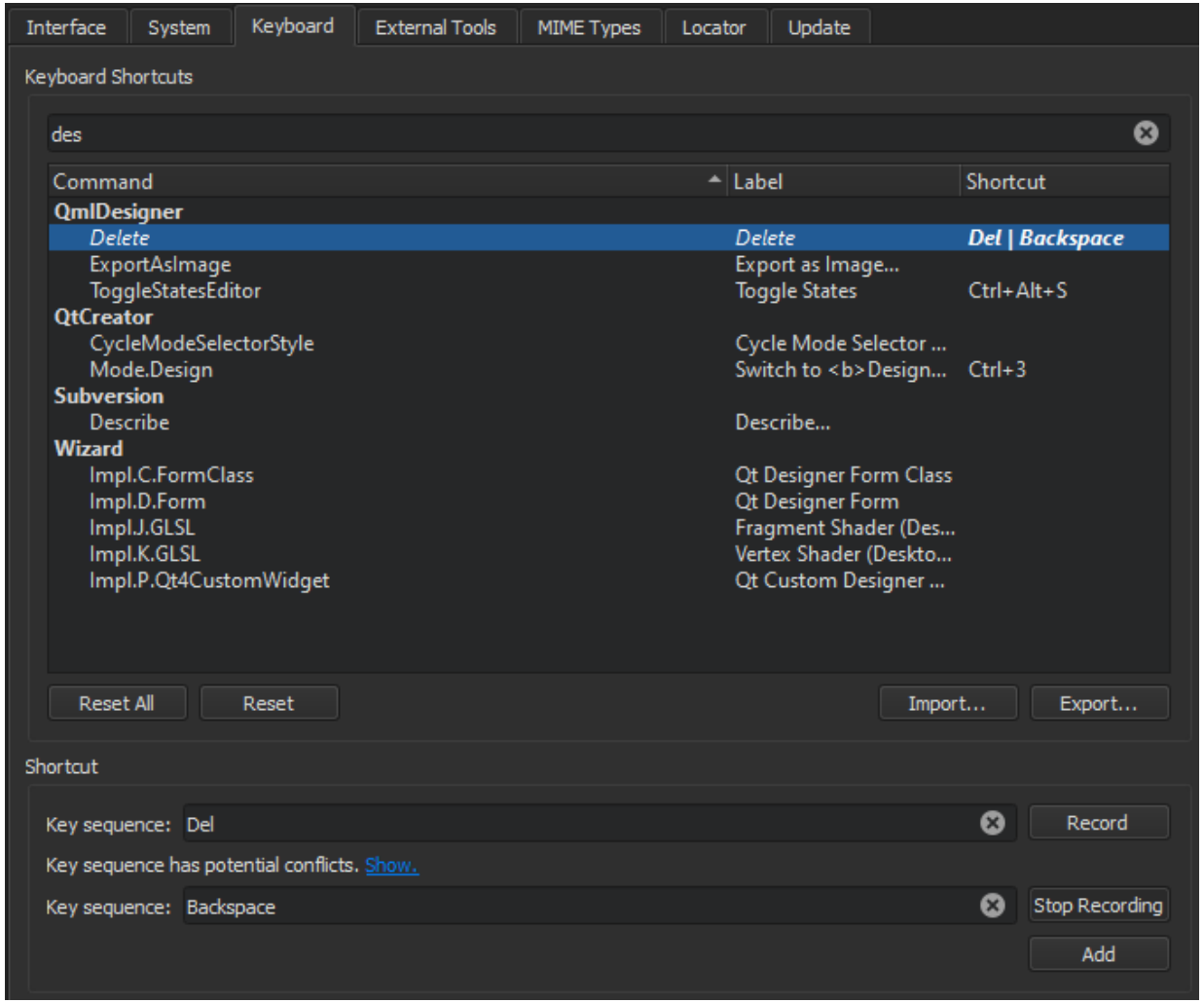




Heuristic #6: Recognition Rather Than Recall

It's easier for us to recognize something we've seen before than to recall it from scratch

Power to the People! (and the Pros)

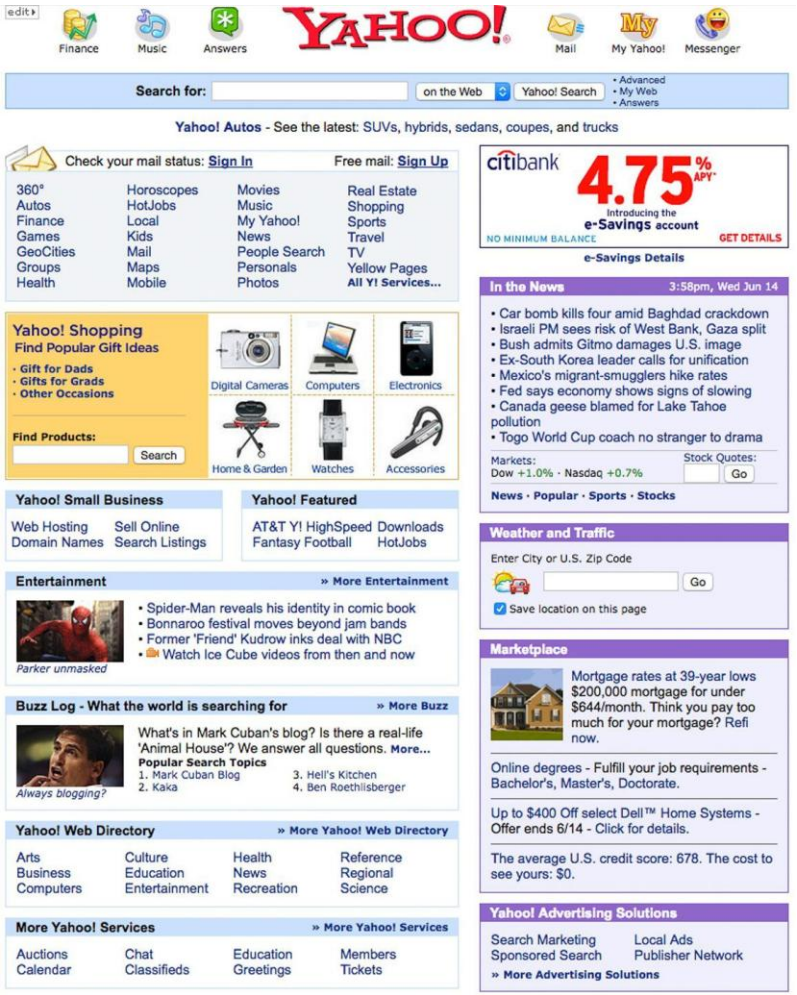


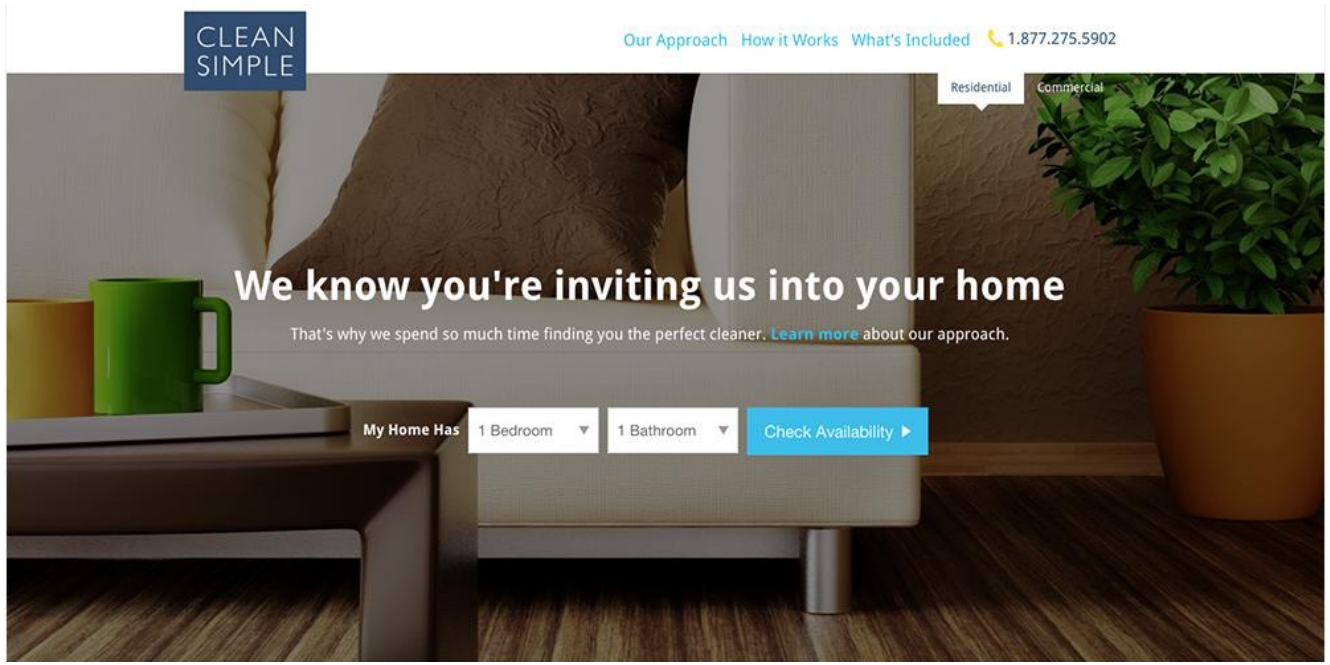


#7: Flexibility and Efficiency of Use

A great system caters to both inexperienced users and experienced experts."

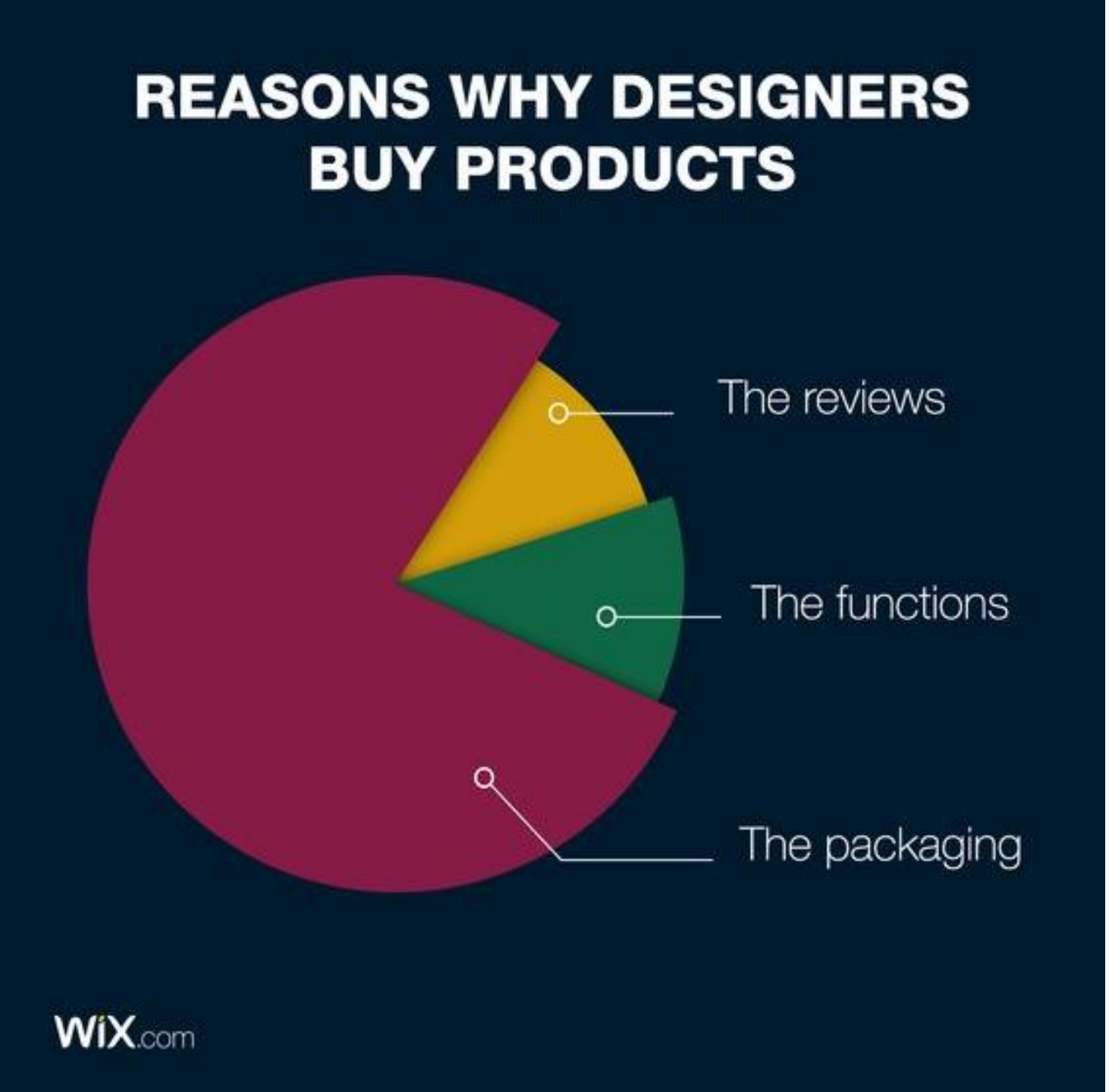
Which page lets you accomplish your goal faster?





Heuristic #8: Aesthetic and Minimalist Design

Every extra piece of information competes with the relevant information.



Relevance when buying something

maybe for all of us...

Theme 4: The Co-pilot - Trust & Uncertainty

Designing for AI: 5 extra rules

<- Which AI assistant would you keep? ->



Clippy

Microsoft Office, 1997

“It looks like you’re writing a letter. Would you like help?”

Pops up while you work.



ChatGPT

2022

“Ask anything”

Waits until you ask.

Plot twist: who pays when the chatbot is wrong?

- Nov. 2022: Jake Moffatt's grandmother dies. He needs to fly from Vancouver to Toronto for the funeral.
- He asks the chatbot on Air Canada's website about bereavement fares (discounts for travelling after a death in the family).
- Chatbot: buy a full-price ticket now, ask for the discount within 90 days.
- Air Canada: no discount after the trip.
- **In court, Air Canada argues the chatbot is "a separate legal entity that is responsible for its own actions".**

Tribunal, 2024

No. The chatbot is part of your website. Air Canada pays 812 Canadian dollars.

Lesson

An AI mistake is a product mistake. Your company owns it.

Moffatt v. Air Canada, 2024 BCCRT 149 (Civil Resolution Tribunal of British Columbia, Feb. 2024)

5 rules for AI features

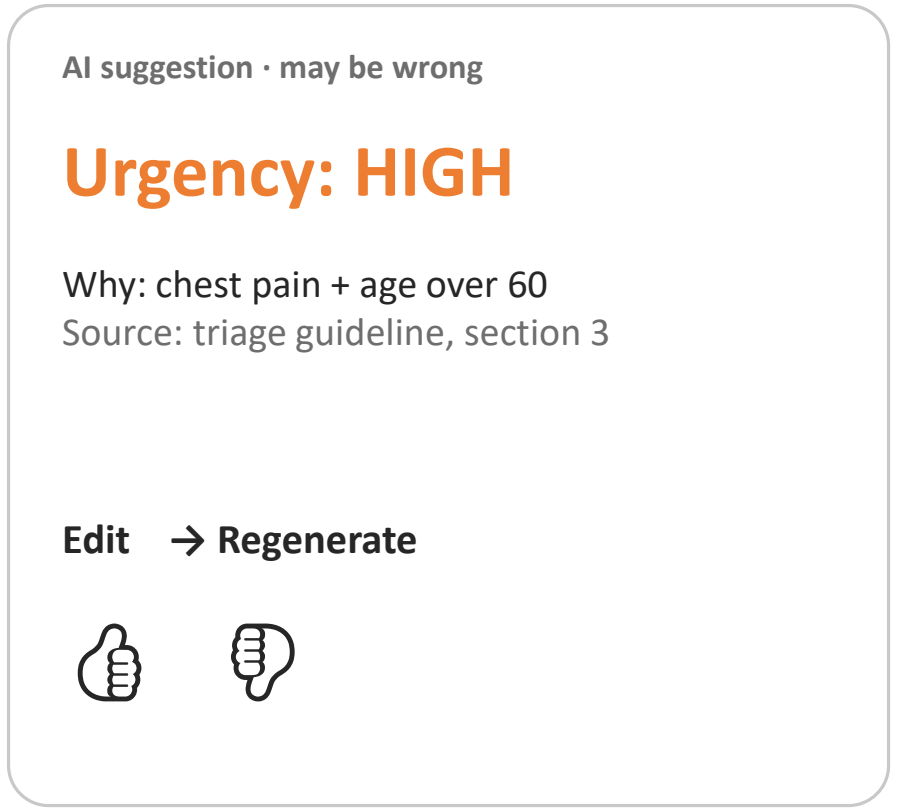
1. **Say what it can and can’t do**
   Set expectations: “AI suggestion, may be wrong.”

2. **Show that it’s working**
   Streaming text, progress: “Analyzing your image…”

3. **Let the user fix it**
   Edit, regenerate, undo. The human has the last word.

4. **Explain why**
   Show the reasons or the sources behind the answer.

5. **Ask for feedback**
   Thumbs up / down shows you where the AI fails.



Example: a digital triage app

After Amershi et al., Guidelines for Human-AI Interaction (Microsoft, CHI 2019) · Google PAIR, People + AI Guidebook

Build with AI, verify with heuristics

1 Specify

Your user stories from SPECS.md. What must the AI screen show: waiting, result, error?

You

2 Build with AI

Generate the screens: v0, Lovable, Bolt, Claude, Copilot…

AI + you

3 Verify

Walk through the 10 heuristics and the 5 AI rules. Fix what fails.

You

AI draws a nice UI in minutes.

It has never met your user.

Milestone 2: Walking skeleton (due in 2 weeks · RO: week 5 · EN: week 6)

Theme: The whole product runs end to end. The AI is still fake.

Tasks:

- **UI:** the main screens of your MVP (web or mobile), covering the user stories from SPECS.md.
- **Mocked AI:** the function that will call the model already exists, with the real input and output format. For now it returns hardcoded or random answers. The UI shows every AI state: waiting, result, error.
- **Mocked data:** hardcoded, in-memory lists. No database yet.

Rule: you may generate the UI with AI. Then check it against the 10 heuristics and the 5 AI rules and fix what fails.

Deliverable: code in your team repository + at least 4 screenshots in the **UI_ss/** folder, including the AI result and the AI waiting or error state.

References

- Nielsen Norman Group: 10 usability heuristics
- Nielsen Norman Group: You are not the user
- CareerFoundry: UX vs UI design
- Video (YouTube)
- Amershi et al., Guidelines for Human-AI Interaction (CHI 2019)
- Google PAIR, People + AI Guidebook
- Moffatt v. Air Canada, 2024 BCCRT 149
- Reuters: ChatGPT sets record for fastest-growing user base (2023)